

Data Dependent Branch Profiles

The first chart shows the breakdown of dynamic conditional branches, excluding always-taken and never-taken ones. The profiling information is collected during the last 10-30M instructions of the training traces. The predictor used is equivalent to TAGE_SC.

Legend:

LOAD_DEP0 – Branch is **not** load dependent.

LOAD_DEP1 – Branch depends on **1** load instruction.

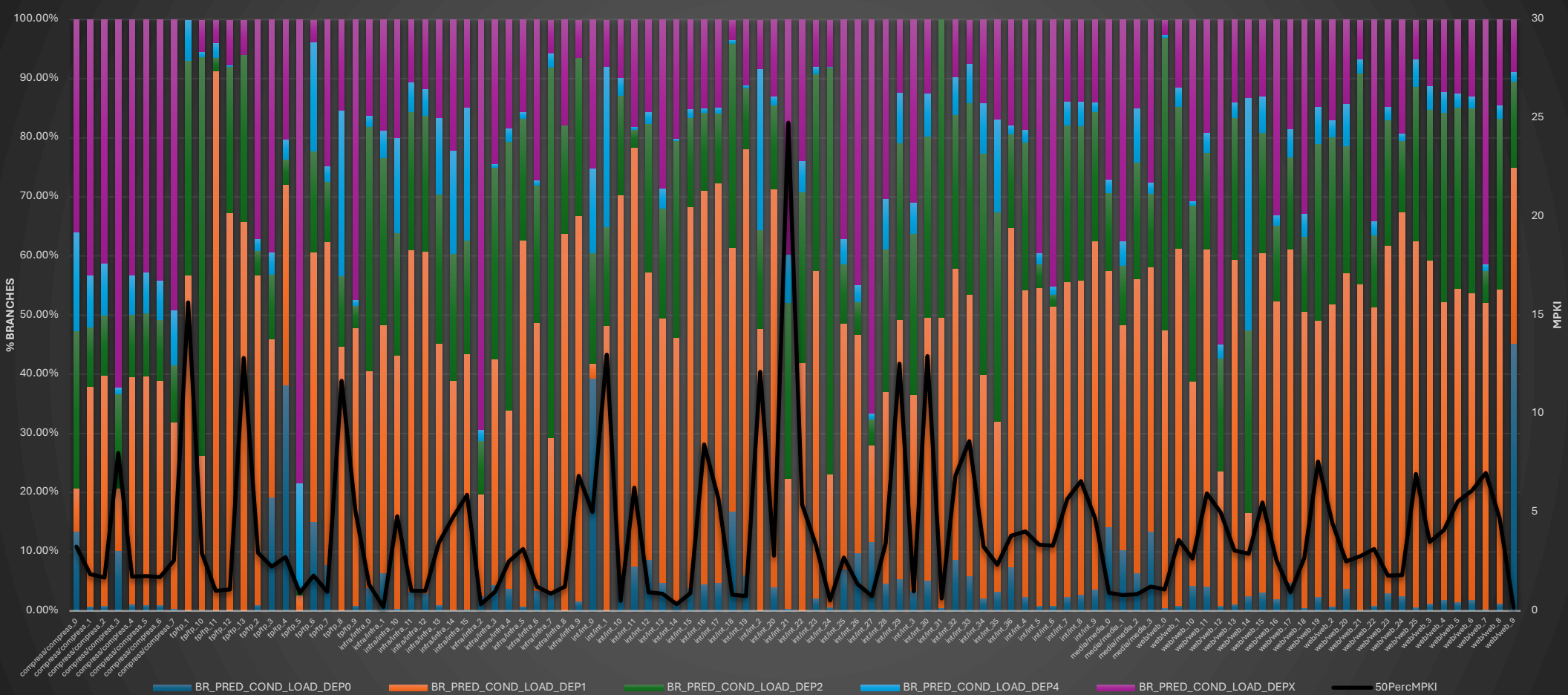
LOAD_DEP2 – Branch depends on **2** load instructions.

LOAD_DEP4 – Branch was dependent on **3 or 4** load instructions.

LOAD_DEPX – Branch either depends on 5+ load instructions, **or** the number of load instructions feeding the branch is not fixed during the execution of the trace.

For convenience, we include the conditional branch MPKIs (post warmup) for CBP2025 **base** predictor – note the black line uses the secondary y-axis.

Breakdown of Dynamic Conditional Branches:



The second chart shows the corresponding breakdown of dynamic conditional branch **mispredictions**.

Breakdown of Dynamic Conditional Branch Mispredictions:

